为 Web 服务配置 SAML 身份验证

ONTAP 9

NetApp
March 17, 2022

This PDF was generated from https://docs.netapp.com/zh-cn/ontap/system-admin/configure-saml-authentication-task.html on March 17, 2022. Always check docs.netapp.com for the latest.

目录

为 Web 服务配置 SAML 身份验证 1
配置 SAML 身份验证 1
禁用 SAML 身份验证 3
对 SAML 配置问题进行故障排除 3

为 Web 服务配置 SAML 身份验证

配置 SAML 身份验证

从 ONTAP 9.3 开始，您可以为 Web 服务配置安全断言标记语言（ SAML ）身份验证。配置并启用 SAML 身份验证后，用户将通过外部身份提供程序（ Identity Provider ， IdP ）进行身份验证，而不是 Active Directory 和 LDAP 等目录服务提供程序进行身份验证。

您需要什么？ **#8217**；将需要什么

- 您必须已为 SAML 身份验证配置 IdP 。
- 您必须具有 IdP URI 。

关于此任务

- SAML 身份验证仅适用于 `http` 和 `ontapi` 应用程序。

  以下 Web 服务使用 `http` 和 `ontapi` 应用程序：服务处理器基础架构， ONTAP API 或 System Manager 。

- SAML 身份验证仅适用于访问管理 SVM 。

步骤

1. 创建 SAML 配置，以便 ONTAP 可以访问 IdP 元数据：

   ` * security SAML -sp create -idP-uri *idp_uri _ -sp-host _ontap_host_name**`

   `IdP_uri` 是可从中下载 IdP 元数据的 IdP 主机的 FTP 或 HTTP 地址。

   `ontap_host_name` 是 SAML 服务提供程序主机的主机名或 IP 地址，在本例中为 ONTAP 系统。默认情况下，使用集群管理 LIF 的 IP 地址。

   您可以选择提供 ONTAP 服务器证书信息。默认情况下，使用 ONTAP Web 服务器证书信息。

```
cluster_12::> security saml-sp create -idp-uri
https://scspr0235321001.gdl.englab.netapp.com/idp/shibboleth -verify
-metadata-server false

Warning: This restarts the web server. Any HTTP/S connections that are
active
         will be disrupted.
Do you want to continue? {y|n}: y
[Job 179] Job succeeded: Access the SAML SP metadata using the URL:
https://10.63.56.150/saml-sp/Metadata

Configure the IdP and Data ONTAP users for the same directory server
domain to ensure that users are the same for different authentication
methods. See the "security login show" command for the Data ONTAP user
configuration.
```

此时将显示用于访问 ONTAP 主机元数据的 URL 。

2. 在 IdP 主机中，使用 ONTAP 主机元数据配置 IdP 。

   有关配置 IdP 的详细信息，请参见 IdP 文档。

3. 启用 SAML 配置：

   ` * security SAML -sp modify -is-enabled true*`

   系统会自动为访问 `http` 或 `ontapi` 应用程序的任何现有用户配置 SAML 身份验证。

4. 如果要在配置 SAML 后为 `http` 或 `ontapi` 应用程序创建用户，请指定 SAML 作为新用户的身份验证方法。

   a. 使用 SAML 身份验证为新用户创建登录方法： + ` * security login create -user-or-group-name *user_name* -application [http ： ontapi] -authentication-method SAML -vserver *svm_name**`

   ```
   cluster_12::> security login create -user-or-group-name admin1
   -application http -authentication-method saml -vserver  cluster_12
   ```

   b. 验证是否已创建此用户条目：

   ` * 安全登录显示 *`

```
cluster_12::> security login show

Vserver: cluster_12
                                                                    Second
User/Group                  Authentication                 Acct
Authentication
Name            Application Method        Role Name        Locked Method
-------------- ----------- ------------- ---------------- ------
--------------
admin           console     password      admin            no     none
admin           http        password      admin            no     none
admin           http        saml          admin            -      none
admin           ontapi      password      admin            no     none
admin           ontapi      saml          admin            -      none
admin           service-processor
                            password      admin            no     none
admin           ssh         password      admin            no     none
admin1          http        password      backup           no     none
**admin1          http        saml          backup           -
none**
```

"ONTAP 9 命令"

禁用 SAML 身份验证

如果要停止使用外部身份提供程序（ IdP ）对 Web 用户进行身份验证，则可以禁用 SAML 身份验证。禁用 SAML 身份验证后，将使用已配置的目录服务提供程序（例如 Active Directory 和 LDAP ）进行身份验证。

您必须从控制台登录。

步骤

1. 禁用 SAML 身份验证

   ` * security SAML -sp modify -is-enabled false*`

2. 如果您不想再使用 SAML 身份验证或要修改 IdP ，请删除 SAML 配置：

   ` * 安全 SAML -sp delete*`

对 SAML 配置问题进行故障排除

如果配置安全断言标记语言（ SAML ）身份验证失败，您可以手动修复 SAML 配置失败的每个节点并从故障中恢复。在修复过程中， Web 服务器将重新启动，并且任何活动的

HTTP 连接或 HTTPS 连接将中断。

配置 SAML 身份验证时， ONTAP 会按节点应用 SAML 配置。启用 SAML 身份验证后，如果存在配置问题，ONTAP 会自动尝试修复每个节点。如果任何节点上的 SAML 配置出现问题，您可以禁用 SAML 身份验证，然后重新启用 SAML 身份验证。有时，即使重新启用 SAML 身份验证， SAML 配置也无法应用于一个或多个节点。您可以确定 SAML 配置失败的节点，然后手动修复该节点。

步骤

1. 登录到高级权限级别：

   ` * 设置 -privilege advanced*`

2. 确定 SAML 配置失败的节点：

   ` * security SAML -sp status show -instance*`

   ```
   cluster_12::*> security saml-sp status show -instance

                            Node: node1
                  Update Status: config-success
                 Database Epoch: 9
    Database Transaction Count: 997
                    Error Text:
   SAML Service Provider Enabled: false
          ID of SAML Config Job: 179

                            Node: node2
                  Update Status: config-failed
                 Database Epoch: 9
    Database Transaction Count: 997
                    Error Text: SAML job failed, Reason: Internal error.
   Failed to receive the SAML IDP Metadata file.
   SAML Service Provider Enabled: false
          ID of SAML Config Job: 180
   2 entries were displayed.
   ```

3. 修复故障节点上的 SAML 配置：

   ` * security SAML -sp repair -node *node_name**`

```
cluster_12::*> security saml-sp repair -node node2

Warning: This restarts the web server. Any HTTP/S connections that are
active
         will be disrupted.
Do you want to continue? {y|n}: y
[Job 181] Job is running.
[Job 181] Job success.
```

Web 服务器将重新启动，并且任何活动的 HTTP 连接或 HTTPS 连接将中断。

4. 验证是否已在所有节点上成功配置 SAML：

` * security SAML -sp status show -instance*`

```
cluster_12::*> security saml-sp status show -instance

                          Node: node1
                Update Status: config-success
               Database Epoch: 9
   Database Transaction Count: 997
                   Error Text:
SAML Service Provider Enabled: false
        ID of SAML Config Job: 179

                          Node: node2
                Update Status: **config-success**
               Database Epoch: 9
   Database Transaction Count: 997
                   Error Text:
SAML Service Provider Enabled: false
        ID of SAML Config Job: 180
2 entries were displayed.
```

Copyright Information

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.